

SunRey Access

Governed Access to Productive Capacity

From ownership to governed economic access in an automated economy. A rights-, capacity-, policy- and solvency-bounded architecture for turning participation into time-limited use of real productive capacity.

CORE DESIGN INVARIANT

Access is neither money nor a third token. It is a governed non-cash entitlement to use defined productive capacity, while token state, Access entitlement state and fiat settlement state remain separate.

Technical / Economic White Paper

September 2026 | SunRey

Conceptual architecture. This publication intentionally omits production allocation coefficients, internal funding assumptions and provider-specific commercial implementation details. Regulated payment, consumer-protection and related capabilities remain subject to jurisdiction-specific legal basis, partner coverage and operational approval.

Executive Abstract

SunRey Access is designed as a new economic layer between participation and consumption. Rather than asking a user to own every productive asset they may need, Access allocates governed rights to use real capacity: a vehicle-day, a room-night, an admission, a unit of compute, a ride, a quantity of energy, or eventually a robot-hour. The unit of economic meaning is not a speculative token balance. It is an enforceable, policy-bounded entitlement to a specific class of useful capacity. [1][2]

The architecture is deliberately separated from money. SunRey Coin and MoonRey Coin may influence eligibility or allocation, but Access itself is not a bank deposit, stablecoin, cash balance or guaranteed redemption promise. A provider can continue to receive conventional fiat while the user experiences a governed Access benefit. This allows an economic-access layer to coexist with today's merchant economy instead of requiring every provider to adopt a new settlement asset. [1]

The result is a three-state system. Token state records participation inputs. Access entitlement state records non-cash rights, reservations, consumption, restoration and expiration. Fiat funding and settlement state records actual monetary commitments, user contributions, provider payments and refunds. Keeping those authorities distinct is the foundation for solvency, auditability, consumer protection and future regulatory reasoning. [1]

CENTRAL CLAIM

The long-run differentiator is not a new payment instrument. It is a programmable rights system that converts governed participation into bounded economic capability without collapsing rights, tokens and money into one balance.

From Ownership to Economic Access

Economic access can become a first-class right without becoming a third currency.

Industrial economies have traditionally linked capability to ownership or disposable income: own a vehicle, own a computer, lease space, buy a ticket, purchase energy, or pay for a service each time it is used. As AI, robotics and infrastructure become more autonomous, the economically relevant question can shift from “Who owns the asset?” toward “Who may use the productive capacity, for what purpose, for how long, and under what constraints?”

SunRey Access is built around that shift. It treats productive capacity as something that can be discovered, classified, allocated, reserved, consumed, restored after cancellation, and audited. A user does not need a legal ownership interest in a hotel, GPU cluster, vehicle fleet or future robotic system in order to receive a defined right to use a portion of its available output.

This is especially important inside SunRey’s dual-economy model. The human economic layer expresses intent, preferences, permissions and participation; the productive layer supplies compute, machines, infrastructure and output. Access is the governed bridge that converts participation in those systems into real-world capability. [2][3]

Access Is Not Money

The most important design decision is negative: Access is intentionally not defined as money. That choice prevents the entitlement system from silently inheriting the semantics of deposits, stored value, stablecoins, cash equivalents or unconditional token redemption.

An Access unit is category-specific and capacity-specific. A room-night is not interchangeable with a GPU-hour merely because both are valuable. Each category can carry its own geography, expiry, reservation rules, eligibility constraints, funding conditions and fulfillment requirements. This makes the system closer to programmable economic rights than to a general-purpose medium of exchange.

SunRey and MoonRey participation may influence how scarce Access capacity is allocated, but the entitlement does not promise a permanent fiat conversion rate and does not require token liquidation at the point of use. The system can therefore avoid representing an Access right as a disguised claim against a token treasury. [1]

NON-NEGOTIABLE CLASSIFICATION

Access Entitlement != Cash != Bank Deposit != Stablecoin != Guaranteed Fiat Redemption. Participation can influence allocation, but actual use remains bounded by capacity, policy and available program funding.

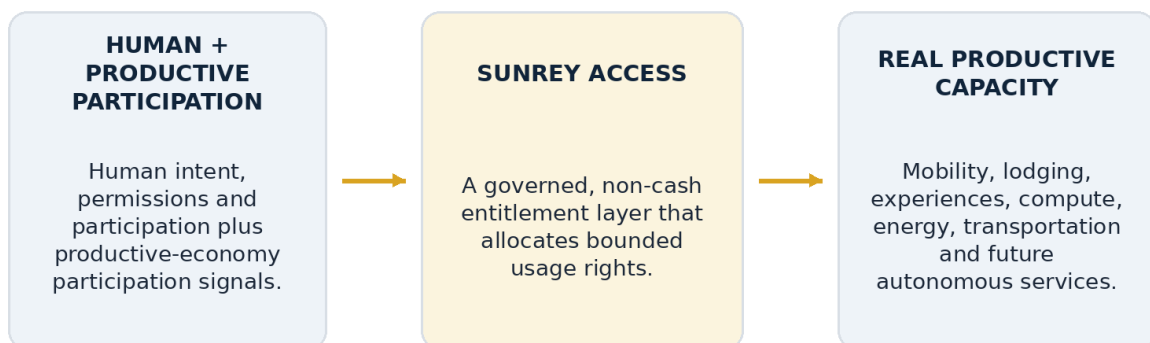
Access Inside the Personal Economic Operating System

SunRey's broader architecture organizes consumer economic life around three reinforcing experiences: participation in the SunRey and MoonRey economies, AI-assisted financial optimization through Grow My Money, and governed Access to productive capacity. The intended result is a Personal Economic Operating System in which conventional financial capital, human economic capital, productive economic capital and economic access can be coordinated without being collapsed into one accounting category. [2]

Within this model, Access is the capability layer. A user may hold money, participate in human-information or productive networks, receive recommendations from Financial Agents, and separately hold rights to use specific capacity. These are related economic resources, but they retain different authorities and legal meanings.

This distinction also protects the architecture from a common design error: assuming that all economic value must be expressed as a tradable token. Access allows some value to remain a right to use, consume or receive a service. That can be economically meaningful even when the right is non-transferable, expiring, category-bounded or dependent on an external provider.

Access as the Bridge from Participation to Capability



The user experiences an economic capability; the provider can still experience ordinary commerce.

Figure 1. Access functions as a governed bridge between participation signals and real productive capacity. The provider experience can remain ordinary commerce.

What Is Productive Capacity?

Productive capacity is the measurable ability of an economic system to provide useful output. In SunRey's architecture, capacity is broader than tokenized asset ownership. A hotel room can produce a room-night of lodging; a GPU can produce compute-hours or inference; a vehicle can produce mobility; an energy system can produce or deliver kilowatt-hours; a future robot can produce service-hours.

This capacity perspective matters because ownership and utility are not identical. Many valuable assets are under-utilized for part of their lifecycle, while many users need temporary capability rather than permanent possession. A rights layer can match those two facts without requiring the user to own the underlying productive asset.

The Access architecture therefore represents capacity using operational dimensions such as category, unit, availability window, geography, fulfillment constraints, provider capability, price components, policy state and evidence of consumption. It is a practical bridge between productive-economy measurement and consumer use. [1][3]

CAPACITY QUESTION

The system asks: What useful output is actually available, where and when can it be used, what rules govern it, and can the entitlement be honored without exceeding capacity or funding?

A Multi-Domain Access Vocabulary

Access requires a common economic vocabulary without pretending that every capacity unit is identical. The architecture therefore uses category-specific units that preserve the operational meaning of the underlying service.

A category can be divisible or indivisible, reservable or instantly consumable, refundable or non-refundable, local or remotely delivered. These characteristics affect allocation, reservation, evidence and reconciliation. The public model can remain stable even as the provider network changes underneath it.

PUBLICATION BOUNDARY

This paper describes the category model and control architecture, but does not publish provider-specific commercial terms, production inventory assumptions or proprietary capacity sourcing strategies.

Capacity domain	Representative Access unit	Illustrative use
Mobility	vehicle-hour / vehicle-day / ride	Rental, shared mobility, future autonomous transport
Lodging	room-night	Hotel or short-stay inventory
Experiences	admission	Events, tours, recreation, entertainment
Food	meal / food basket	Eligible meal or grocery benefit
AI / Compute	GPU-hour / inference unit	Metered compute or AI service consumption
Energy	kWh	Charging or future electricity access
Transportation	ride / trip	Transit and transport
Robotics	robot-hour	Future autonomous service capacity

Entitlements: Rights Rather Than Balances

An Access entitlement is best understood as a rights object. It can state that an eligible user may reserve or consume a defined quantity of capacity under specified conditions. The entitlement may be time-limited, category-bounded, geography-bounded, non-transferable, subject to availability, and revocable under defined program rules.

This design permits the system to represent economic capability with greater precision than a generic wallet balance. A user can see what they are entitled to do - not merely how many abstract units they possess. The right can move through explicit lifecycle states such as allocated, available, reserved, consumed, restored, expired or cancelled.

Because the entitlement is not itself money, refund and cancellation behavior can restore rights separately from refunding fiat. If a booking is cancelled before consumption, the system can release a capacity reservation and restore the entitlement while a separate payment workflow returns eligible monetary amounts. That separation is essential to deterministic reconciliation. [1]

RIGHTS MODEL

The economically meaningful object is: who may use what capacity, for which purpose, during which period, under which policy, and with what fulfillment evidence.

Three Independent States

The Access design maintains three authoritative state domains. Each answers a different question and no one domain is permitted to silently substitute for another.

Token state answers what SunRey and MoonRey participation exists and how that participation evolved. Access entitlement state answers what non-cash rights are available, reserved, consumed, restored or expired. Fiat funding and settlement state answers what money has actually been committed, authorized, paid, refunded or reserved. [1]

This separation prevents a token balance from being treated as a merchant liability, prevents an entitlement from being treated as a deposit, and prevents a fiat payment event from silently changing token ownership. It also provides a clean audit trail when a transaction partially succeeds or fails.

Three-State Separation

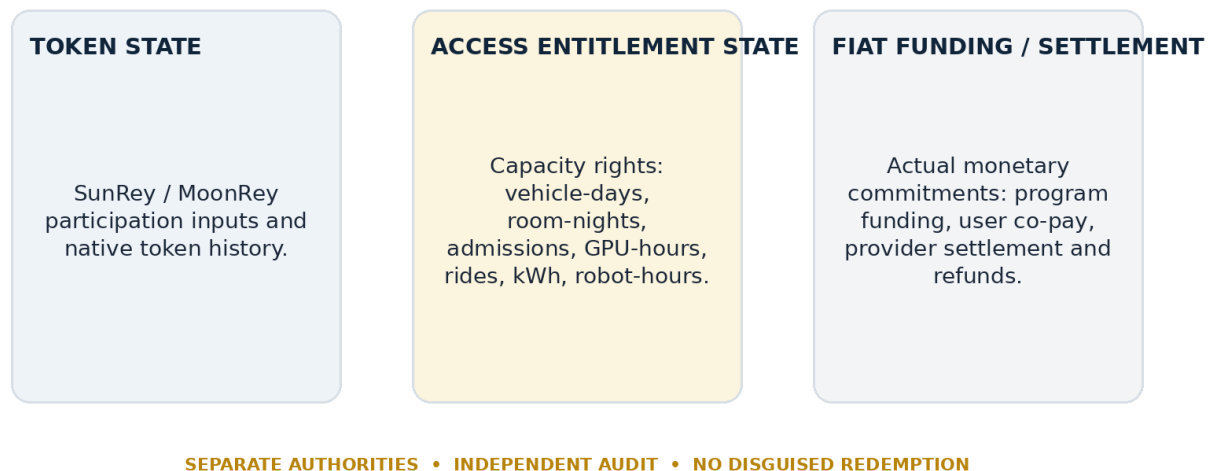


Figure 2. Token state, Access rights and fiat settlement are separate authorities with separate accounting and audit semantics.

Canonical Authority Boundaries

Access depends on multiple systems, but the architecture assigns each one a narrow authority. External discovery providers may report availability. An Access policy engine may determine eligibility. The entitlement ledger may reserve and consume rights. A payment system may authorize the user contribution. A provider may confirm fulfillment. None of these facts, by itself, is allowed to become the canonical truth for the other domains.

The same principle applies to AI. A Financial Agent can discover an Access opportunity and explain why it may fit a user's goals. It cannot independently mint an entitlement, create capacity, approve regulated payment, override policy, or declare a provider transaction complete.

This is the broader SunRey doctrine applied to economic access: external data informs; policy authorizes; canonical ledgers determine financial state; and provider evidence proves fulfillment. [1][4]

Governance and Authority Stack

CONSUMER INTENT	Select, reserve, confirm, cancel
ACCESS POLICY	Eligibility, category, geography, limits
ENTITLEMENT AUTHORITY	Reserve, consume, restore, expire
COMPLIANCE + RISK	Jurisdiction, consumer protection, transaction controls
FIAT LEDGER + PAYMENT RAIL	Authorize user contribution and provider settlement
EVIDENCE + RECONCILIATION	Receipts, provider state, refunds, audit

Figure 3. A layered authority model keeps consumer intent, Access policy, entitlement state, compliance, fiat settlement and evidence distinct.

Allocation Is a Governance Problem

When productive capacity is scarce, allocation becomes an economic-governance problem rather than a simple pricing problem. SunRey's design uses participation signals as inputs to a governed allocation process, but preserves the rule that no allocation can exceed actual eligible capacity.

The architecture is intended to resist snapshot gaming and linear concentration. Participation is measured over time rather than at a single instant, and the allocation transform introduces diminishing marginal influence as participation increases. The design can also reward engagement across both the human and productive sides of the dual economy.

The important public principle is not a particular coefficient. It is that the allocation function is policy-versioned, deterministic, auditable and capacity-bounded. Allocation rules can evolve through governance without pretending that those rules are market prices or guaranteed exchange rates. [1][2]

PUBLICATION BOUNDARY

Production coefficients, reference balances, category-specific weights and internal thresholds are intentionally omitted. Those parameters are implementation policy and potential intellectual property, not required to understand the architecture.

Time-Weighted Participation

An instantaneous wallet snapshot is easy to manipulate: acquire a position immediately before an allocation event, receive a disproportionate benefit, then exit. The Access design instead measures participation over a defined observation window. This reduces the importance of a single balance at a single moment and makes participation history economically relevant.

Time weighting also creates a clearer governance surface. A policy can define the observation period, eligible assets or participation classes, exclusions, cut-off time and version. The allocation snapshot can then be finalized for a period so that user rights do not continuously fluctuate with every block or market movement.

The result is a separation between a dynamic participation signal and a stable entitlement period. Once an entitlement is issued for a defined cycle, the user can make plans around it; subsequent participation changes affect future allocations rather than retroactively rewriting existing rights. [1]

DESIGN OBJECTIVE

Measure durable participation without converting Access into a continuously floating token derivative.

Diminishing Returns and Broad Participation

If Access rights scaled linearly with token balances, scarce capacity could be captured by the largest holders. The source architecture therefore uses a concave allocation transform: additional participation continues to matter, but each incremental unit contributes less influence than the previous one.

This is not presented as a normative claim that every user should receive equal capacity. It is an engineering and governance choice that can make a limited resource pool more broadly distributable while still recognizing participation. It also reduces the economic incentive to concentrate holdings solely to capture Access allocation.

Because the transform is a governed policy rather than a token price, the system can evaluate its effects empirically. Allocation concentration, utilization, unmet demand, user satisfaction, provider availability and program solvency can all be monitored before changing policy for the next cycle.

ECONOMIC PRINCIPLE

Scarce capacity should be allocated by an explicit rule, not accidentally by the side effects of a wallet balance.

Capacity-Bounded Allocation

The allocation engine must satisfy a hard physical constraint: the sum of rights issued for a capacity category cannot exceed the amount of capacity that the program can actually honor. The entitlement system therefore treats available capacity as an upper bound rather than as an aspirational target.

For divisible capacity, deterministic rounding can preserve fractional units. For whole-unit categories, the allocation process must resolve indivisibility without creating phantom inventory. In either case, the finalized allocation is versioned and auditable so a later policy change does not alter historical rights.

Capacity bounds also connect economic design to operational reality. Discovery inventory is not the same thing as funded, fulfillable capacity. A search result may show that a service exists; an entitlement can be honored only when policy, provider capability, available capacity, funding and transaction controls all align. [1]

Capacity-Bounded Allocation



Figure 4. Participation signals are transformed through governed rules, eligibility and hard capacity limits before an entitlement is finalized.

The Entitlement Lifecycle

A robust Access system needs more than an “available balance.” Each right moves through a state machine that reflects whether it can still be used and whether an external commitment exists.

A typical lifecycle is: allocated -> available -> reserved -> consumed. Cancellation or provider failure can move a reserved right into a restored state when policy permits. Expiration closes an unused entitlement after its defined period. Administrative or compliance actions may pause the ability to create new reservations without erasing the historical record.

Reservation is especially important. Before a provider booking is finalized, the system should atomically reserve both the entitlement and any required program funding so the same right or same funding is not committed twice. If the transaction fails, resources are released only after remote state is understood.

STATE DISCIPLINE

A right can be visible to a user yet temporarily unavailable for funded redemption if provider capacity or program funding is exhausted. Visibility and fulfillability are not the same state.

Funding Is a Separate Constraint

A non-cash entitlement can still create a real monetary obligation when SunRey subsidizes or funds a provider purchase. That means Access needs a funding architecture separate from entitlement allocation.

The program may be funded by multiple approved sources, such as program treasury, subscriptions, sponsor commitments, employer or government programs, provider economics or other governed sources. These sources can carry restrictions by category, geography, user cohort or program. The architecture must preserve those restrictions through reservation, settlement and refund.

The crucial point is that allocated rights do not imply unlimited funding. A user may hold an entitlement while a particular funded fulfillment path is temporarily unavailable. The system should surface that condition rather than allowing funding to go negative or pretending that an entitlement is equivalent to cash. [1]

SOLVENCY PRINCIPLE

Economic rights may be allocated only within the program's ability to fulfill them. Entitlement issuance and financial funding are related, but they are not the same ledger.

Solvency, Reserves and Commitments

Access introduces a treasury problem: the platform can promise rights only to the extent that the rights can be funded or fulfilled. The system therefore tracks available funding, pending commitments, refund exposure and risk reserves before approving new subsidized transactions.

This creates a form of entitlement solvency distinct from banking solvency. The objective is not to represent Access as a deposit liability. It is to ensure that outstanding program commitments do not exceed the resources designated to satisfy them.

Program governance can set category caps, transaction caps, user limits, geographic limits and emergency pause controls. A treasury pause should stop new funded commitments while preserving the ability to service existing bookings, refunds, history and reconciliation. This is a narrow safety boundary, not a discretionary ability to rewrite completed rights.

CONTROL OBJECTIVE

No new funded commitment should be accepted when eligible available funding is insufficient. Existing commitments remain serviceable and auditable.

Quote, Coverage and User Contribution

Access does not require every eligible transaction to be fully subsidized. A coverage engine can classify which parts of a firm provider quote are eligible, determine how much of the user's remaining entitlement and program funding can be applied, and calculate any remaining user contribution.

This requires cost-component classification. The underlying service cost may be eligible while upgrades, refundable deposits, tips, late fees, damage exposure or unrelated incidentals remain the user's responsibility. A clear classification model makes the customer experience explainable and keeps the Access treasury from inheriting open-ended contingent liabilities.

The coverage decision is computed before financial execution. It should be policy-versioned, quote-aware and bounded by available entitlement, eligible funding and transaction limits. If the provider price changes, the system requotes and recalculates rather than silently increasing the user's contribution. [1]

CONSUMER PROTECTION

The user should know what Access covers, what they must pay, what remains contingent, and what happens if the price or booking state changes.

Merchant Settlement Can Remain Ordinary Fiat

One of the most practical features of the architecture is that Access does not require merchants to accept SunRey Coin or MoonRey Coin. The provider can receive an ordinary fiat settlement amount through an approved payment rail while SunRey handles the internal combination of program contribution and user contribution. [1][2]

This abstraction separates the user benefit from merchant adoption. A hotel, transport operator, compute service or experience provider does not need to understand SunRey's token economy in order to fulfill an Access transaction. From the provider's perspective, the transaction can look like conventional commerce with ordinary booking and payment semantics.

At launch, the architecture can operate without forced token sale, burn, transfer or liquidation at checkout. That reduces coupling between Access fulfillment and digital-asset market liquidity, while preserving the possibility of more native productive-capacity settlement models in later phases if regulation, custody and provider participation mature.

LAUNCH PRINCIPLE

The user experiences an Access benefit; the provider can still receive conventional settlement. Token adoption is not a prerequisite for useful economic access.

Capability-Driven Provider Abstraction

Access should not hard-code its economic model around any single merchant, travel platform, compute vendor or mobility network. The provider domain is therefore capability-driven: discovery, firm quote, reservation, booking, fulfillment, cancellation, refund, settlement and reconciliation are represented as interfaces that different providers can implement.

This abstraction makes it possible to combine open discovery data, self-service transactional services, commercial partnerships and future direct capacity contributors. It also provides a clean replacement and fallback mechanism when one provider is unavailable or lacks a required geography or capability.

The architecture keeps provider health and provider terms as inputs rather than as financial authority. External systems can prove that capacity exists or a booking completed, but they do not determine user entitlement, internal program funding or canonical fiat balance state. [1]

PUBLICATION BOUNDARY

Specific provider names, onboarding assumptions, commercial rates and routing preferences are intentionally omitted from this public paper.

Discovery Is Not Authority

The Access experience depends on real-world data: availability, schedules, locations, capacity, pricing, provider status and fulfillment evidence. SunRey's external-data architecture routes these observations through a provider registry, reliability and security controls, provenance tracking, cache policy and a deterministic trust layer before they become canonical product inputs. [1]

This matters because a search result is not a financial fact. Discovery may be stale, incomplete or non-transactional. The Access system must distinguish "this opportunity appears to exist" from "this capacity is reservable at a firm price" and from "this capacity has been funded and fulfilled."

The trust layer can evaluate freshness, source authority, corroboration, outliers and provider risk without allowing external data to bypass SunRey policy. This allows Access to use a broad ecosystem of real-world information while preserving a single internal control model.

AUTHORITY RULE

External data informs. SunRey policy authorizes. Entitlement and fiat ledgers establish internal state. Provider evidence proves remote fulfillment.

A Deterministic Transaction State Machine

A complete Access transaction crosses multiple failure domains: provider inventory, entitlement state, program funding, user payment, compliance controls and remote booking state. Treating this as a single API call would create duplicate bookings, double-spent entitlements and unreconciled money when networks fail.

The architecture instead uses an explicit state machine. The user discovers an opportunity, receives a firm quote, sees Access coverage and any co-pay, reserves the entitlement and funding, authorizes their contribution, completes provider booking and settlement, receives fulfillment, and then the system reconciles all authoritative states.

Unknown outcomes are first-class states. If a provider appears to have booked but the response is lost, the system reconciles before retrying. If payment succeeds and booking fails, the system compensates or refunds according to the known remote state. This is safer than assuming every timeout means failure. [1]

Access Transaction State Machine



Unknown remote state is reconciled before retry. Idempotency prevents duplicate bookings, payments

PROVIDER STATE + FIAT LEDGER STATE + ENTITLEMENT STATE

Figure 5. Access uses a deterministic state machine across discovery, reservation, payment, fulfillment and reconciliation rather than treating checkout as one opaque action.

Reservation, Booking and Fulfillment Evidence

Reservation is where an abstract right becomes a concrete economic commitment. The Access system must bind a user entitlement to a particular quote, provider, quantity, time window and policy version before the external booking proceeds.

A successful booking is not yet the same as consumption. Some capacity is fulfilled immediately; other capacity is consumed later. The architecture therefore retains provider reservation identifiers, booking state and evidence references until the service is delivered or the transaction is cancelled.

Fulfillment evidence can vary by category. Compute may provide metered usage; an experience may provide an admission confirmation; lodging may provide stay completion; mobility may provide rental or trip completion. The architecture does not require every provider to expose identical evidence, but it does require enough evidence to reconcile the right that was consumed.

EVIDENCE PRINCIPLE

The system should be able to reconstruct which right was reserved, what money moved, what the provider reported, what was actually fulfilled, and what state remains afterward.

Refunds, Restoration and Partial Failure

Access refunds are inherently multi-state. A provider refund can change fiat state, while a cancelled or failed service can also restore all or part of an entitlement. Those two actions must be coordinated without assuming they are identical.

The architecture uses compensation and reconciliation rather than destructive rollback. If a booking is cancelled, the system records the cancellation, verifies the provider outcome, determines the eligible monetary refund, releases or restores the entitlement according to policy, and preserves the complete history.

This design is particularly important when a remote system is uncertain. The platform should not restore a right while a provider booking remains active, and it should not duplicate a refund because the same webhook arrived twice. Idempotent events and ownership checks protect against these races. [1]

FAILURE RULE

Never retry an unknown booking or payment state blindly. Reconcile first, then apply one deterministic transition.

Operational Invariants

The architecture can be summarized by a small set of hard invariants. They are stronger than user-interface checks because they define states the system is never allowed to enter.

- Consumed Access units cannot exceed the user's allocated units.
- Reserved plus consumed units cannot exceed the allocation.
- Committed Access funding cannot exceed eligible available funding.
- Available funding cannot become negative.
- A refund cannot exceed the monetary amount actually captured.
- Provider settlement must equal approved funding sources for that transaction.
- The same entitlement, booking, payment or refund event cannot be applied twice.
- Client-supplied price, coverage, funding source or entitlement identity is never accepted as financial authority.

WHY INVARIANTS MATTER

A compelling Access experience can fail economically if it permits double-spend, negative funding, duplicate booking or unexplained settlement. The protocol must make those states impossible or explicitly recoverable.

Security, Privacy and Minimum Necessary Data

Access crosses financial and provider boundaries, so data minimization is part of the economic design. A provider needs the information required to fulfill a booking; it does not need a user's complete financial profile, Human Information Network history, Vault contents, token holdings or unrelated personal data.

Payment credentials and provider secrets should remain inside controlled payment and provider boundaries rather than appearing in the consumer application, ordinary logs, receipts or general event streams. Webhooks must be authenticated, mutation endpoints must be idempotent, and provider transport should block arbitrary proxying into internal networks or cloud metadata services. [1]

The result is an Access system that can use rich SunRey context to determine eligibility internally while disclosing only the minimum data required externally. That distinction is important if Access later expands into sensitive domains such as health-adjacent services, public benefits or employer programs.

PRIVACY PRINCIPLE

Internal eligibility context is not provider fulfillment data. Only the minimum necessary booking information should cross the provider boundary.

What the User Actually Experiences

The consumer interface should hide most of the underlying architecture while preserving transparency about rights and obligations. A user can discover eligible opportunities, see available entitlements, request a quote, understand how much Access applies, review any required co-pay, confirm a reservation, cancel when permitted, and see history and remaining capacity.

The interface should clearly distinguish an entitlement from cash. It should show category, unit, expiry, reservation status, coverage conditions and any remaining user responsibility. If a funded redemption is temporarily unavailable, the product should say so rather than representing the entitlement as a failed payment balance.

Receipts should explain both economic layers: what service was obtained and what financial contribution was made. The Action Center can surface pending confirmations, price changes, expirations, refund states and reconciliation issues without exposing internal provider complexity.

PRODUCT PRINCIPLE

Simple experience, explicit rights: “You have access to X” is more accurate than pretending every benefit is a cash balance.

Policy, Jurisdiction and Consumer Protection

Access can look deceptively simple at the consumer layer, but live deployment can involve payments, merchant contracting, consumer disclosures, refunds, tax treatment, digital assets and program accounting. The architecture therefore makes jurisdiction and policy part of the transaction decision.

The Compliance Kernel remains the deterministic authority for applicable transaction controls. External providers and AI systems cannot bypass it. Jurisdiction-specific policy can determine whether a category is available, what disclosures apply, what payment rail may be used, whether a user or provider is eligible, and which legal entity is responsible. [1][2]

This is also why Access is intentionally classified as a non-cash right rather than casually marketed as redeemable money. The system should make the economic promise it can actually honor and maintain records sufficient for consumer support, accounting, treasury review and dispute resolution.

REGULATORY BOUNDARY

Software architecture can encode controls and evidence. It does not by itself establish legal permission to operate a live payments, digital-asset, benefits or consumer-finance program.

Treasury Limits, Policy Versioning and Kill Switches

Access governance controls two scarce resources: productive capacity and program funding. Both should be represented through versioned policies rather than hidden operational assumptions.

Governance can define the allocation observation window, transformation family, category eligibility, capacity sources, geography, entitlement duration, user caps, transaction caps, funding limits, reserve requirements and provider health thresholds. Changes apply prospectively to new allocation or transaction cycles and are recorded so historical decisions remain reconstructable.

Emergency controls should be narrow. A treasury kill switch can stop new funded commitments if exposure exceeds a limit. A provider quarantine can stop new bookings when a remote system becomes unreliable. Neither control should erase existing bookings or prevent users from seeing history, cancelling where permitted, receiving refunds, or completing reconciliation.

GOVERNANCE OBJECTIVE

Change policy without changing history. Pause new risk without abandoning existing obligations.

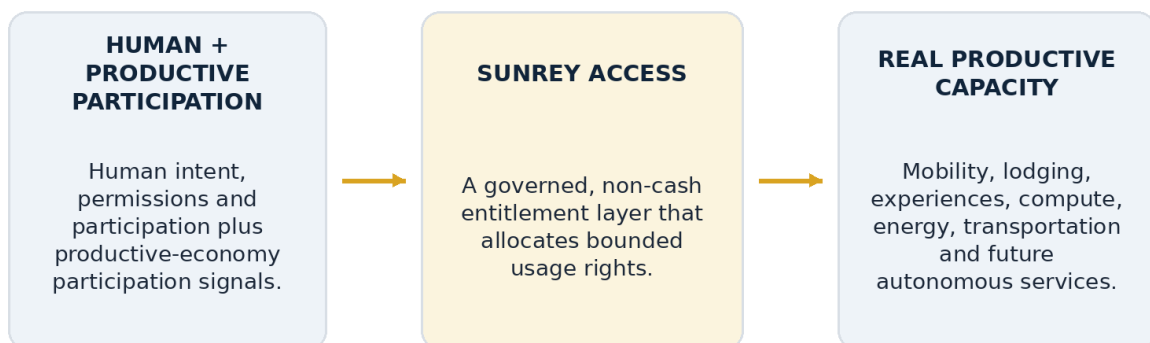
Access Connects Human and Productive Economic Layers

SunRey's dual-economy thesis separates human economic value from autonomous productive value. Humans express goals, preferences, permissions, knowledge and demand; automated systems supply compute, machines, energy, infrastructure and output. Access is the mechanism that converts participation in those layers into a right to use real capacity. [2][3]

This creates a different relationship between digital participation and physical utility. SunRey Coin can coordinate human-economic participation. MoonRey Coin can coordinate productive-economic participation. Access does not merge those assets; it reads governed participation signals from them and produces a separate entitlement state.

The architecture therefore supports a coherent economic loop without claiming that a token is literally backed by every service it can help unlock. Participation, entitlement and fulfillment are connected through policy and evidence rather than through a simplistic one-token-equals-one-service redemption promise.

Access as the Bridge from Participation to Capability



The user experiences an economic capability; the provider can still experien

Figure 6. Access turns governed participation into a separate layer of economic capability while preserving the independence of the underlying human, productive and settlement systems.

From Productive Output to Personal Economic Capability

A core SunRey proposition is that economic well-being should be measured by more than financial balances. Personal Economic Capability can include financial capital, human economic capital, productive economic participation and the actual ability to use goods and services. [2]

Access makes the final category explicit. A user who has time-limited rights to transportation, compute or lodging possesses a real economic capability even if that right is not a transferable asset and cannot be withdrawn as cash.

This framing may become increasingly relevant in a highly automated economy. If productive systems can generate more output with less direct human labor, a useful economic architecture needs mechanisms that connect people to that capacity without requiring proportional ownership of every machine or infrastructure asset. SunRey Access is one proposed mechanism for that bridge.

MACROECONOMIC THESIS

Ownership remains important, but access rights can become an independent dimension of economic capability in an economy with abundant, automated productive capacity.

Institutional and Sponsored Access

The entitlement model is not limited to a direct-to-consumer subsidy. Because funding and entitlement state are separate, the same architecture can support employer programs, sponsor-funded benefits, public-sector capacity programs, membership programs, educational compute grants or other institutionally funded Access pools.

Each funding source can preserve its own category and eligibility restrictions. An employer might fund transportation or compute but not entertainment. A sponsor might support a geography or cohort. A public program might require statutory eligibility and reporting. The entitlement layer can remain common while policy and source-of-funds controls vary.

This institutional path is particularly useful because providers can still be paid through ordinary settlement rails. An institution does not need to create a new token economy to deliver a programmable, auditable right to capacity. The architecture can instead use familiar funding while gaining stronger entitlement controls.

DESIGN FLEXIBILITY

Separate funding, entitlement and provider layers allow multiple program sponsors to use one Access architecture without commingling their economic rules.

Toward Native Capacity Markets and Machine Commerce

The launch architecture intentionally stops before direct token conversion, provider acceptance of MoonRey, native productive-capacity markets and machine-to-machine token settlement. Those capabilities materially change custody, exchange, payment, treasury and regulatory obligations. [1]

The longer-term architecture nevertheless points toward a deeper productive economy. A machine or infrastructure operator could eventually publish verified capacity, a policy-aware market could discover that capacity, an authorized agent could reserve it, and settlement could occur through a native economic protocol when the legal and operational environment supports it.

Access can therefore serve as a practical consumer bridge today while remaining compatible with a future in which productive-capacity rights become more natively programmable. The important architectural discipline is sequencing: prove non-cash entitlements, provider fulfillment, solvency and reconciliation before coupling the system to more ambitious native settlement.

V1 STOP LINE

Do not use future machine-commerce aspirations to weaken the controls required for today's consumer entitlement and fiat-settlement model.

Where the Architecture May Be Distinctive

The defensible idea is not any single technology. Time-weighted balances, payment rails, booking APIs, ledgers and entitlement systems all have precedents. The differentiated architecture is their integration into a rights-governed economic system that keeps participation, capacity, entitlement, money, policy and evidence as first-class but separate states.

Several design clusters may merit formal intellectual-property review: capacity-bounded entitlement allocation influenced by dual-economy participation; policy-versioned rights that remain distinct from token and fiat state; provider-independent fulfillment with deterministic three-state reconciliation; and an integrated Personal Economic Operating System that coordinates financial capital, human economic value, productive capacity and Access. [2]

Any patentability conclusion would require a dedicated prior-art search and legal claim analysis. The white paper therefore describes a technical thesis rather than asserting that every component is novel in isolation.

NOVELTY THESIS

The potential moat is the economic ontology and control architecture: who may access what productive capacity, under which governed participation and funding rules, with separate rights, money and evidence state.

Risks and Open Research Questions

Access creates important design questions that should be tested rather than assumed away. A high-quality production system needs empirical evidence about allocation fairness, user behavior, utilization, provider economics, funding sustainability and operational failure rates.

- How should capacity be allocated when demand persistently exceeds supply?
- Which time-weighting and diminishing-return transforms produce broad participation without destroying incentives?
- How should category-specific scarcity and geography affect allocation policy?
- How much reserve is required for cancellations, price volatility, refunds and provider uncertainty?
- When should an entitlement be transferable, non-transferable, expiring or revocable?
- How should the system communicate an entitlement that exists but is temporarily not fundable?
- What evidence is sufficient to prove consumption across compute, mobility, lodging, energy and future robotics?
- How can Access expand internationally without turning one global product into an unmanageable regulatory surface?

RESEARCH STANDARD

Treat allocation and program economics as testable policy, not ideology. Publish outcomes, measure concentration and utilization, and version changes explicitly.

A Phased Deployment Model

A disciplined Access rollout should begin narrowly. One geography, one or two capacity categories, at least one production-capable fulfillment path, one approved settlement rail, a restricted cohort and explicit live-versus-simulated labeling are easier to test than an immediate global marketplace. [1]

Before broad launch, the full lifecycle should be exercised: discovery, quote, coverage, user approval, reservation, provider booking, settlement, fulfillment, cancellation, refund, entitlement restoration, unknown-state reconciliation and customer support. Technical readiness is not enough if treasury limits, provider contracts, disclosures or operational runbooks are incomplete.

The architecture's existing V1 stop line is therefore a strength. It allows SunRey to prove the entitlement model and ordinary merchant settlement before introducing direct digital-asset conversion, native capacity trading or machine-to-machine settlement.

LAUNCH DISCIPLINE

Prove one complete rights-to-fulfillment loop before scaling categories, geographies, providers or settlement complexity.

From Assets to Capability

SunRey Access proposes a simple but consequential shift in economic design: represent the right to use productive capacity as a first-class economic object. The right is not a bank balance, not a third token and not an unconditional redemption promise. It is a governed entitlement whose value comes from the real capability it can unlock.

The architecture works because it refuses to collapse distinct authorities. SunRey and MoonRey participation can inform allocation. Access entitlement state records rights. Funding state protects solvency. Fiat ledgers record money. Provider systems prove fulfillment. Policy and compliance determine what is permitted. Reconciliation binds the states together after real-world execution.

In a more automated economy, this separation may provide a practical path from digital participation to tangible utility. People need not own every productive asset to benefit from the capacity around them. They need trustworthy rules that determine what they may use, when they may use it, and whether the system can actually honor the promise.

That is the purpose of SunRey Access: to turn productive capacity into governed economic capability.

FINAL THESIS

From ownership to access. From abstract participation to real capability. From one undifferentiated balance to explicit rights, policy, capacity, funding and evidence.

Glossary

Term	Meaning
Access	A governed non-cash right to use defined productive capacity.
Access entitlement	A versioned rights object representing category-specific capacity available to a user.
Capacity	The measurable ability of a provider, asset or system to deliver useful output.
Coverage	The portion of an eligible provider cost that the Access program is authorized and funded to support.
Entitlement ledger	The canonical state for allocation, reservation, consumption, restoration and expiry of Access rights.
Fiat settlement state	The canonical monetary state for program funding, user contributions, provider settlement and refunds.
Participation signal	A governed input derived from eligible SunRey/MoonRey participation and used in allocation policy.
Provider abstraction	Capability-based interfaces for discovery, quote, reservation, fulfillment, cancellation, refund and reconciliation.
Reconciliation	The process of resolving provider state, fiat ledger state and entitlement state into a consistent outcome.
Time weighting	A method that measures participation over a period rather than from a single instantaneous snapshot.

PUBLIC INTERPRETATION

Nothing in this glossary should be read as a promise of cash redemption, a bank deposit, a guaranteed service, or a statement that every capacity category is live in every jurisdiction.

Source Basis and Publication Boundary

This white paper synthesizes SunRey's existing Access, business-plan, dual-economy and blockchain architecture documents. It is a public-facing architectural treatment rather than a reproduction of internal engineering instructions. Exact production coefficients, internal funding formulas, named provider routing preferences, commercial assumptions and operational credentials have been deliberately excluded.

[1] SunRey. External Data & Access Architecture Program. August 31, 2026. Source for the non-cash classification, three-state separation, capacity categories, time-weighted and diminishing-return allocation design, capacity bounds, settlement architecture, provider abstraction, transaction state machine, operational invariants and V1 stop line.

[2] SunRey. Comprehensive Business Plan. September 2026. Source for the Personal Economic Operating System framing, Access as one of three consumer experiences, productive-capacity interpretation, allocation principles and partner-first regulatory boundaries. Fundraising, valuation, staffing and go-to-market material are not used in this paper.

[3] SunRey. SunRey + MoonRey Dual-Economy Architecture. August 2026. Source for the human-economic / autonomous-productive distinction and the role of productive capacity as a first-class economic object.

[4] SunRey. SunRey Blockchain & Universal Economic Exchange - Technical Architecture and Novel Market Design. August 2026. Source for the broader sovereign economic state-machine model, non-ownership rights, productive-capacity state, policy, provenance and evidence boundaries.

IMPORTANT

This document describes conceptual architecture and economic design. Live capabilities remain subject to technical validation, provider agreements, funding sufficiency, accounting treatment, security review, legal analysis, consumer-protection requirements and jurisdiction-specific authorization.